PLATE	OBJECT	R.A. 1900	DEC. 1900	TEL.	EMUL.	LST START	LST END	H.A. MID	EXP.	OBS.
V-1937-0417	B-17 LOCAL	05 37 20.0	+32 10 00	24-IN REFL.	Agfa Astro	05 47 15.0	05 48 45.0	+00 10 40.0 W	90 S	S.W.

Seeing 2; 5 F; drive contact lost after exposure

Ruth Bell / plate to vault 10:02 P.M.

VALE OBSERVATORY

PHOTOGRAPHIC PLATE STORE

LOCAL SERIES V / JACKET FORM P-4

SERIES / NUMBER: V-1937-0417

DATE-NIGHT: January 16-17, 1937

OBJECT: Bellweather Field B-17 (Vale local provisional)

TELESCOPE: Vale 24-inch photographic reflector

PLATE: Agfa Astro, 13 x 18 cm

EXPOSURE: 90 SEC. / S.W.

ISSUED: R. BELL

RETURNED: 10:02 P.M.

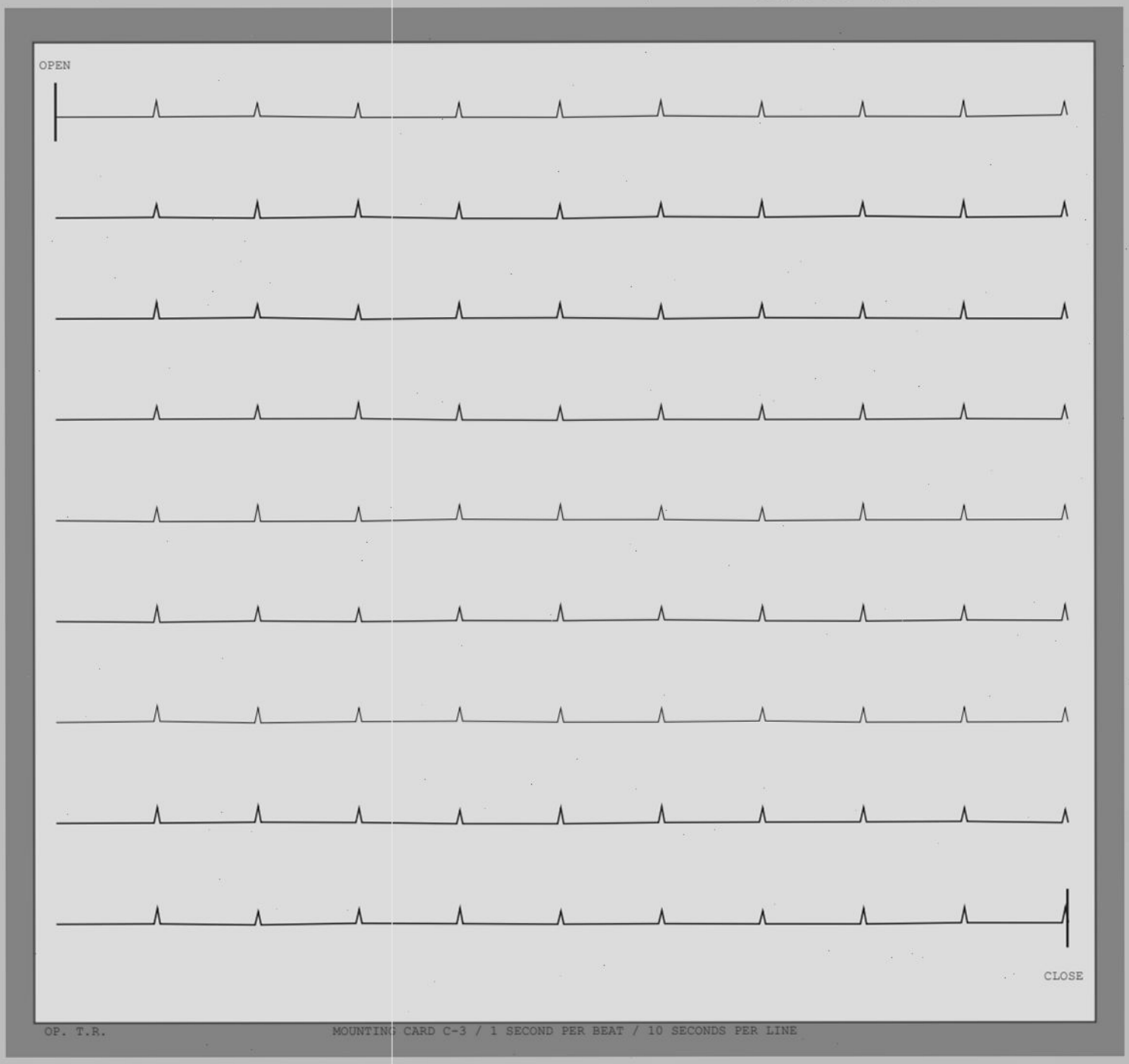
do not clean reverse notes

VALE OBSERVATORY / LOCAL SERIES V / PLATE V-1937-0417

CENTRE 05 37 20.0 +32 10 00 SCALE 60"/MM CARD 17-A

open 05 48 11.8

close 05 49 41.8



mains contact out; beats continue

SIDEREAL STANDARD CORRECTION Vale Observatory / Time Service sheet S-2-1937-016

Clock: Sidereal standard S-2 Date: 16 January 1937

Comparison: U.S. Naval Observatory radio time / 18:00 E.S.T.

Comparison prepared: T. ROOK

Measured result: S-2 FAST 56.8 SEC.

Observation: local-series plate V-1937-0417

Mid-exposure clock reading 05 48 56.8

Convention: add the signed correction algebraically; a fast standard requires a minus sign.

CORRECTION ENTERED BY F. MERCER ... +56.8 seconds

Reduction destination: desk sheet R-17 / plate ledger IV-117

Mercer signature:

DATE / EST	AUTHORITY	CLOCK	FAST	RATE	OP.
JAN 14 18:00	U.S.N.O.	S-2	56.70 S	+0.05 S/D	T.R.
JAN 15 18:00	U.S.N.O.	S-2	56.75 S	+0.05 S/D	T.R.
JAN 16 18:00	U.S.N.O.	S-2	56.80 S	+0.05 S/D	T.R.

S-2 fast; subtract 56.8 s

U.S.N.O. signal sets error; daily difference sets rate

RA - HA - LST REDUCTION Vale Observatory / Reduction desk sheet R-17 / 17 January 1937

Rule: LOCAL SIDEREAL TIME = RIGHT ASCENSION + HOUR ANGLE

DESK RULE EXAMPLE

R.A. 04 10 00.0 + H.A. 00 20 00.0 W = L.S.T. 04 30 00.0

PLATE V-1937-0417

R.A. 05 37 20.0 + H.A. +00 10 40.0 W = L.S.T. 05 48 00.0

Raw S-2 contacts: 05 48 11.8 / 05 49 41.8

Raw S-2 midpoint: 05 48 56.8

Required signed correction: -56.8 seconds

Corrected contacts: 05 47 15.0 / 05 48 45.0

Check: 05 48 56.8 plus (-56.8) = 05 48 00.0

CIVIL TIE — GATEHOUSE CHRONOMETER CARD L-2

Sign separation: 113.6 sidereal sec = 113.3 mean-solar sec

Honest (-) plate close: 10:02:14.0 P.M. E.S.T.

Posted (+) plate close: 10:04:07.3 P.M. E.S.T.

Library entry register: F. MERCER / 10:04 P.M. E.S.T.

Reduced: S. WREN / entered Night Book IV leaf 117

DOME AND CLOCK MAINTENANCE

DATE	SYSTEM	WORK	INITIALS
JAN ~~16~~ 12	S-2 contact	Contact cleaned; unauthorized adjustment	T.R.
JAN 15	rate check	Change under one second for reunion interval	T.R.
JAN 16	dome drive	Motor circuit serviceable before storm	T.R.
JAN 16	chase C	Old weight removed; panel and lock retained	T.R.

TO THE PRESS AND SUBSCRIBING STATIONS

Internal field label: Bellweather Field B-17 (Vale local provisional)

Elements provisional; positions are for finder use only.

Jan. 16, 10 P.M. EST R.A. 05h 37m Decl. +32 10

Estimated magnitude: 8.1
5.8

Copies to: North Notch Press, radio desk, subscriber circular list

L.H.

ENDOWMENT TRANSFER SCHEDULE Vale Observatory / Treasurer copy / T-37-0116-M

RESTRICTED INSTRUMENT ENDOWMENT — CUSTODIAN ADVICE

STATUS: POSTED 16 JANUARY 1937 / LEDGER 41, FOLIO 6

Account ..... Instrument Reserve 41-6

Advice date ..... 16 January 1937 Voucher T-37-0116-M

Opening principal ... $240,000.00

Debit — collateral .. $18,500.00

Balance forward $221,500.00

Payee advice ..... North Notch Equipment Holding Co.

Remittance ..... cashier's draft 0116-C

Authorized ..... F. Mercer, Trustee

Second trustee ..... [blank]

CARBON LETTER - TRUSTEES Vale Observatory / retained desk carbon / draft 2 / 16 January 1937

From: Dr. Alistair Vale, Director

Draft — not issued

To the Trustees and the Astronomical Community:

The first identification on plate V-1911-0086 was Ruth Bell's.

I published the result under my name and permitted the record to stand.

Tonight I intend to restore her credit and submit the endowment books

to an independent audit. The observatory cannot ask reality to remember

only those facts that preserve its founder.

PLATE VAULT APPOINTMENT

Vale Observatory / Director desk carbon / 16 January 1937

FELIX -

PLATE VAULT. TEN O'CLOCK.

BRING TRANSFER SCHEDULE T-37-0116-M.

A. VALE

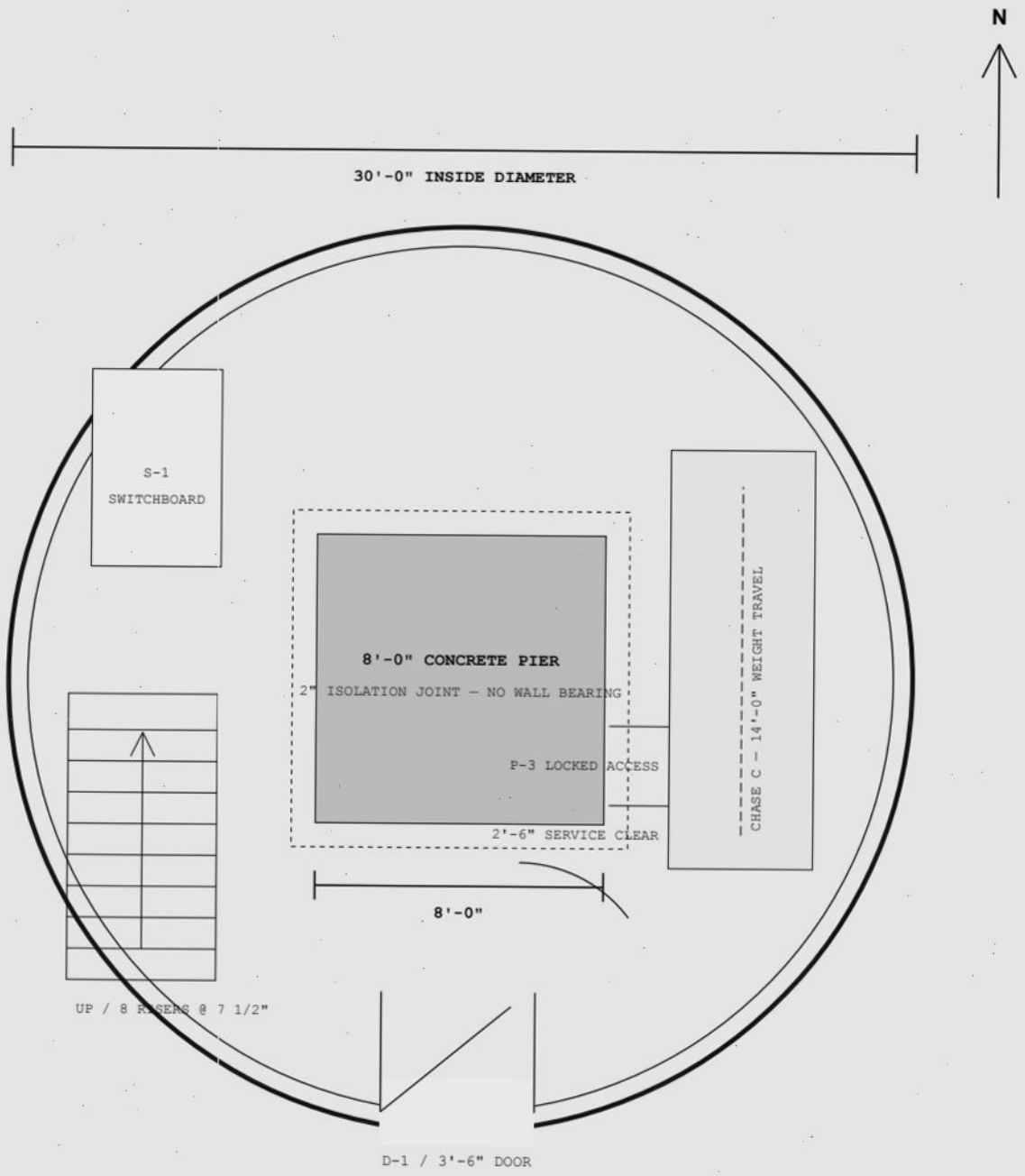
SERVICE KEY REGISTER

Vale Observatory / Plant Book III leaf 23 / dome ground floor / key C

DATE	HOLDER	OUT	IN	PURPOSE
JAN 15	T. ROOK	3:20 P.M.	4:05 P.M.	drive oil
JAN 16	FELIX MERCER	9:52 P.M.	10:18 P.M.	trustee inspection

C panel

DOME GROUND-FLOOR PLAN



0 2 4 6 8 FT

SERVICE RECOVERY NOTE / PLANT BOOK III

17 Jan 10:42 — panel P-3 opened.

Body of Dr. Alistair Vale recovered from chase C. T.R.

VALE OBSERVATORY

DOME GROUND-FLOOR SERVICE PLAN

PROJECT: DRIVE & TIME-SERVICE ALTERATIONS
SCALE: 1/4" = 1'-0"

DRAWN: T.R. 12 JAN 1937
CHECKED: A.V. 15 JAN 1937

DWG. D-4
REV. 1
SHEET 4 OF 6

STORM AND POWER LOG

TIME E.S.T.	SYSTEM	OBSERVATION
8:40 P.M.	PRIVATE SPUR	drifted shut; 18 in. at upper bend
9:31 P.M.	TELEPHONE	overhead pair lost beyond gatehouse
10:03:12 P.M. E.S.T.	MAINS	lights, dome motor, remote displays lost
10:03:14	S-2 REGULATOR	mechanical standard continued
10:05 P.M.	GENERATOR	started; 14 gal.; lights only
ALL NIGHT	PUBLIC ROAD	reported open to valley gate

Night January 16–17

1 Plate V-1937-0416 — comparison field

2 Plate V-1937-0417 — Bellweather Field B-17 (Vale local provisional)

3 Repeat plate 0417

3 Repeat plate 0417

S

A. Vale

VALE OBSERVATORY

PLATE V-1911-0086

DATE: 17 FEB 1911

B-17 field precursor

R. Bell — object identified and measured

ARCHIVE CHARGE: R.B. / 16 JAN 1937

RAKING LIGHT FROM LEFT / 22.5 DEGREES

VALE OBSERVATORY / LOCAL SERIES V / PLATE V-1937-0416

~~probable minor planet~~

S.W. candidate

S.W.

COPY RL-0416-1 / R. BELL / 17 JAN 1937

GATEHOUSE CHRONOMETER CARD L-2

TIME AUTHORITY: U.S. NAVAL OBSERVATORY RADIO TIME

RADIO COMPARISON: 16 JANUARY 1937 / 6:00 P.M. E.S.T.

MEAN-TIME CHRONOMETER: M-1

SIDEREAL STANDARD: S-2

M-1 E.S.T.	S-2 CORRECTED L.S.T.	OP.	REFERENCE
10:02:14.0 P.M. E.S.T.	05 48 45.0	T.R.	PLATE V-1937-0417 / CLOSE

CONVERSION RULE

1 MEAN-SOLAR SECOND = 1.0027379093508 SIDEREAL SECONDS

113.6 SIDEREAL SECONDS / 1.0027379093508 = 113.3 MEAN-SOLAR SECONDS

POSTED (+) CLOSE: 10:04:07.3 P.M. E.S.T.

LIBRARY REGISTER (MINUTE ENTRY): F. MERCER / 10:04 P.M. E.S.T.